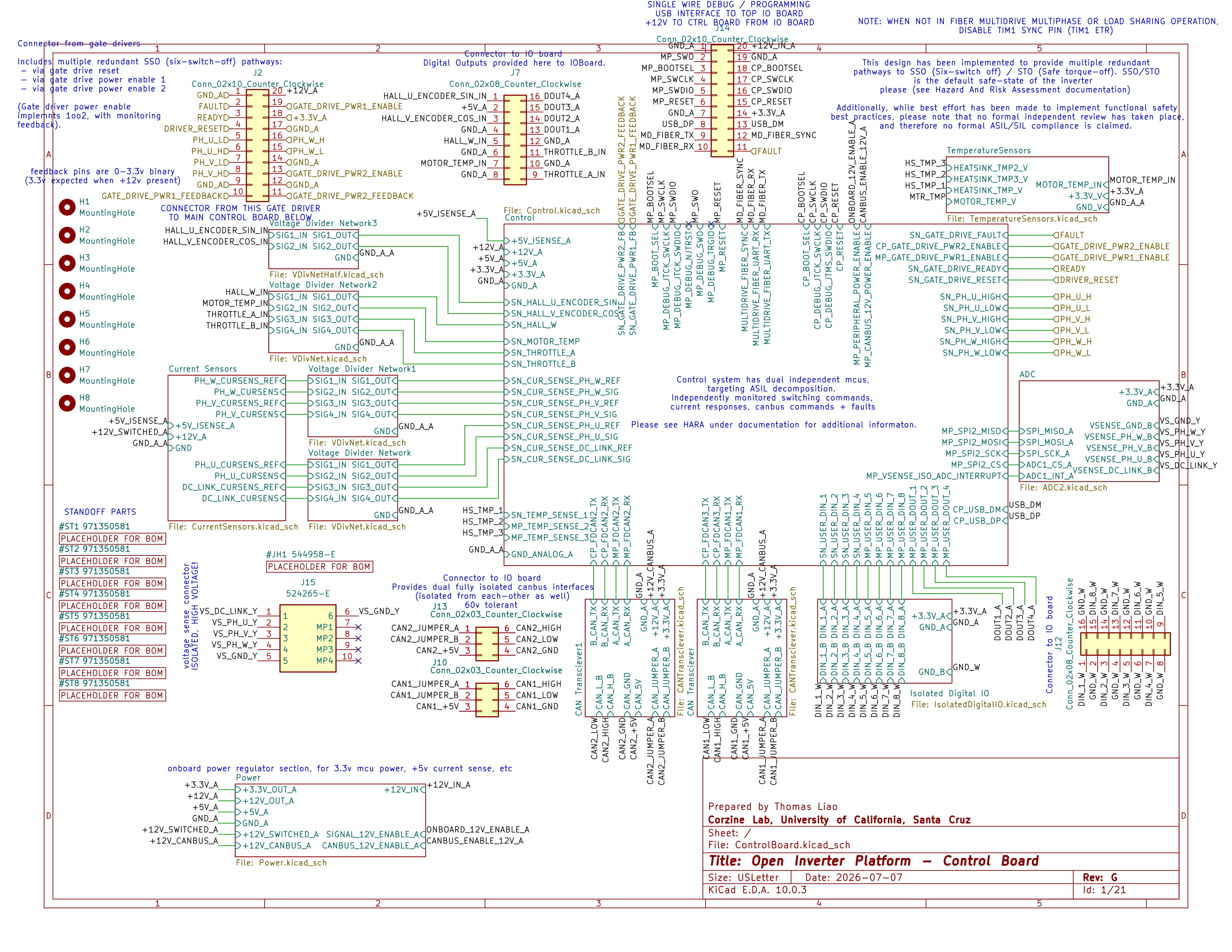
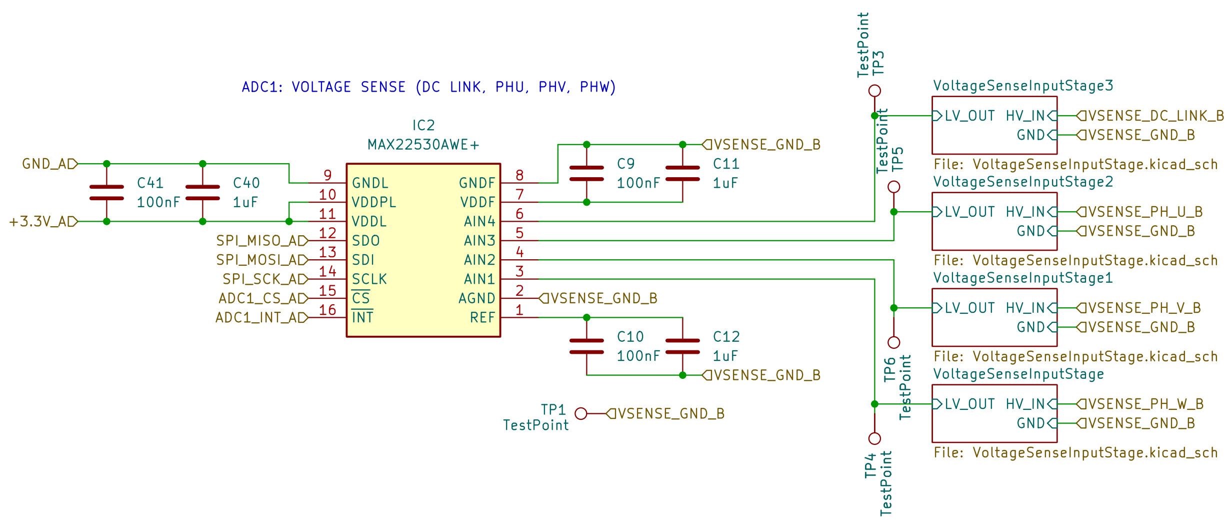


SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION, DISABLE TIM1 SYNC PIN (TIM1 ETR)
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary (3.3v expected when +12v present)
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
File: Control.kicad_sch
File: TemperatureSensors.kicad_sch
File: VDivNetHalf.kicad_sch
File: VDivNet.kicad_sch
File: CurrentSensors.kicad_sch
File: ADC2.kicad_sch
File: CANTransciever.kicad_sch
File: IsolatedDigitalIO.kicad_sch
File: Power.kicad_sch
STANDOFF PARTS
#ST1 971350581 PLACEHOLDER FOR BOM
#ST2 971350581 PLACEHOLDER FOR BOM
#ST3 971350581 PLACEHOLDER FOR BOM
#ST4 971350581 PLACEHOLDER FOR BOM
#ST5 971350581 PLACEHOLDER FOR BOM
#ST6 971350581 PLACEHOLDER FOR BOM
#ST7 971350581 PLACEHOLDER FOR BOM
#ST8 971350581 PLACEHOLDER FOR BOM
#JH1 544958-E PLACEHOLDER FOR BOM
J15 524265-E
voltage sense connector ISOLATED, HIGH VOLTAGE!
Connector to IO board
Provides dual fully isolated canbus interfaces (isolated from each-other as well) 60v tolerant
Connector to IO board
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: Open Inverter Platform — Control Board
Size: USLetter
Date: 2026-07-07
Rev: G
KiCad E.D.A. 10.0.3
Id: 1/21



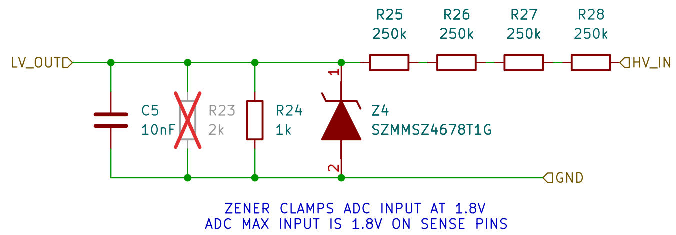

Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /ADC/
File: ADC2.kicad_sch

***Title: Open Inverter Platform – Control Board***

Size: USLetter | Date: 2026-07-07 | **Rev: G**

KiCad E.D.A. 10.0.3 | Id: 4/21




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

***Title: Open Inverter Platform – Control Board***

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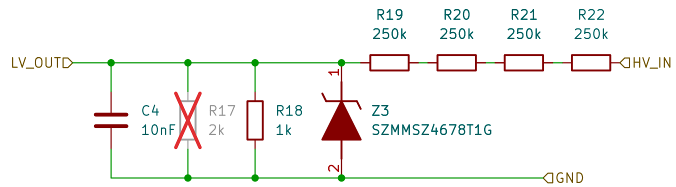
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 2/21 |




ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS


Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**


Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

***Title: Open Inverter Platform - Control Board***

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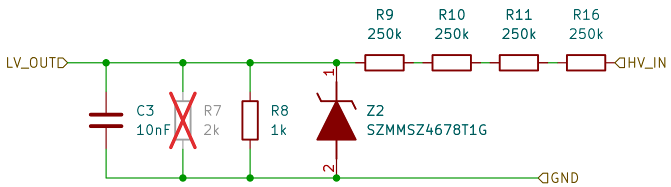
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 3/21 |



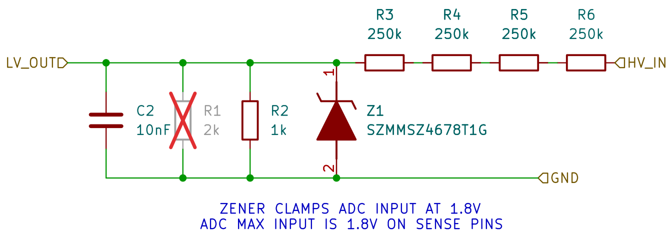

ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

***Title: Open Inverter Platform – Control Board***

Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 5/21




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

***Title: Open Inverter Platform – Control Board***

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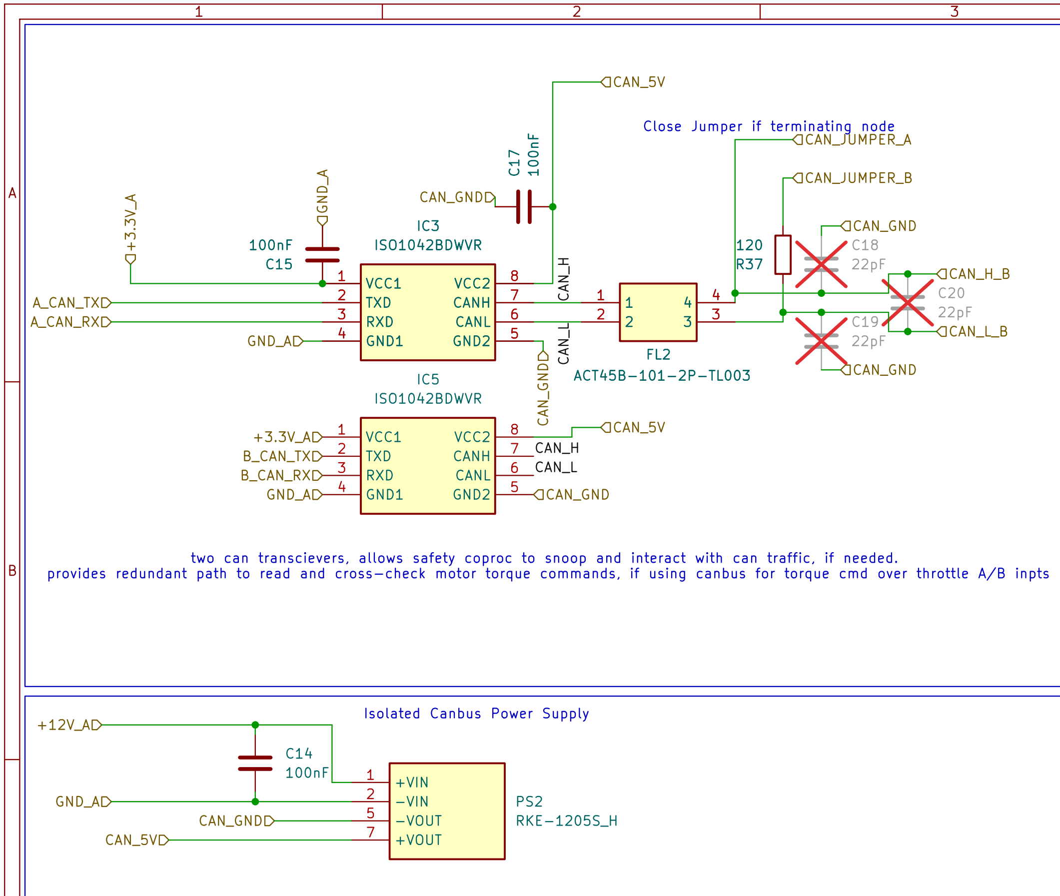
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 6/21 |




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch

***Title: Open Inverter Platform - Control Board***

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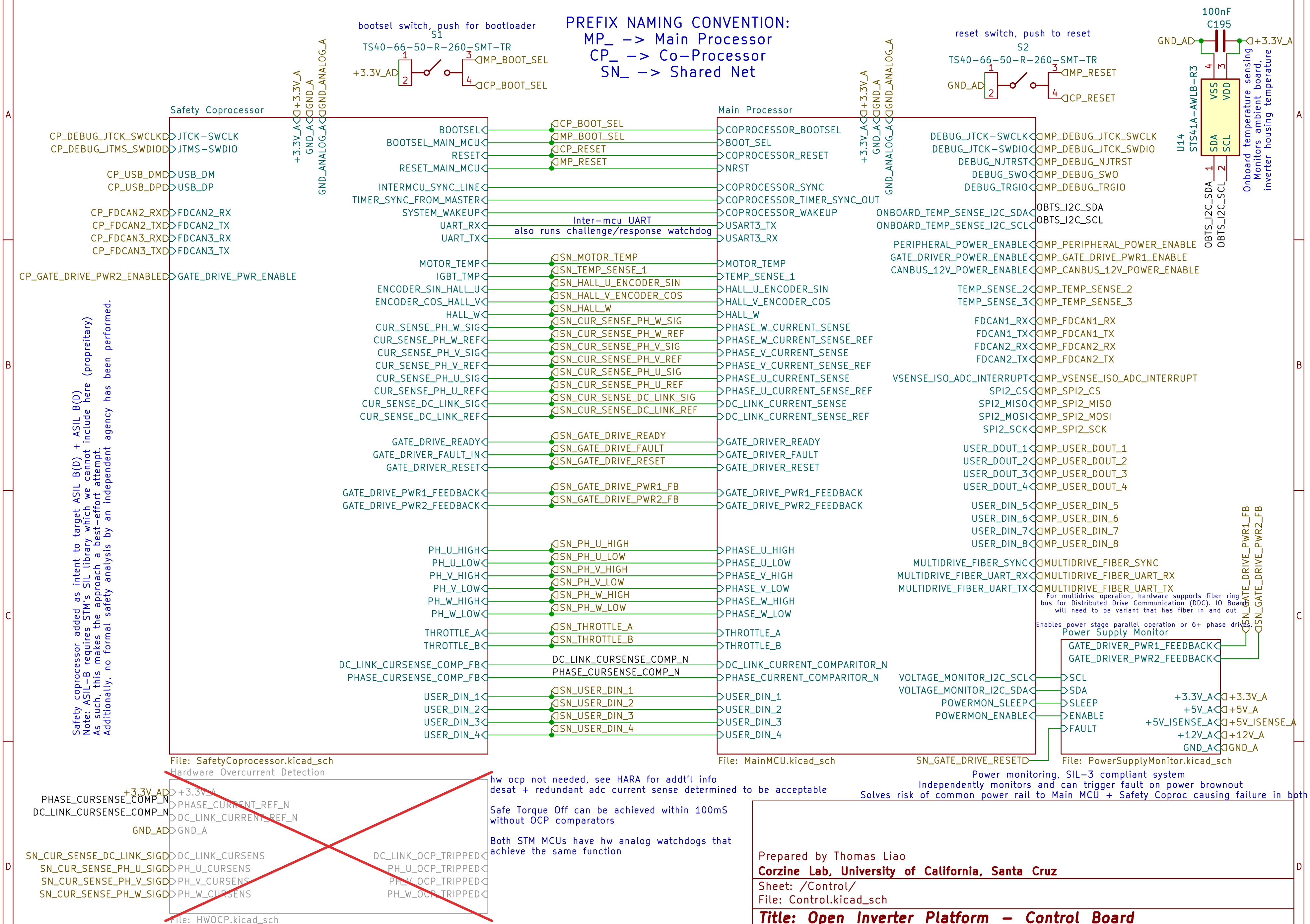
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 7/21 |




Prepared by Thomas Liao

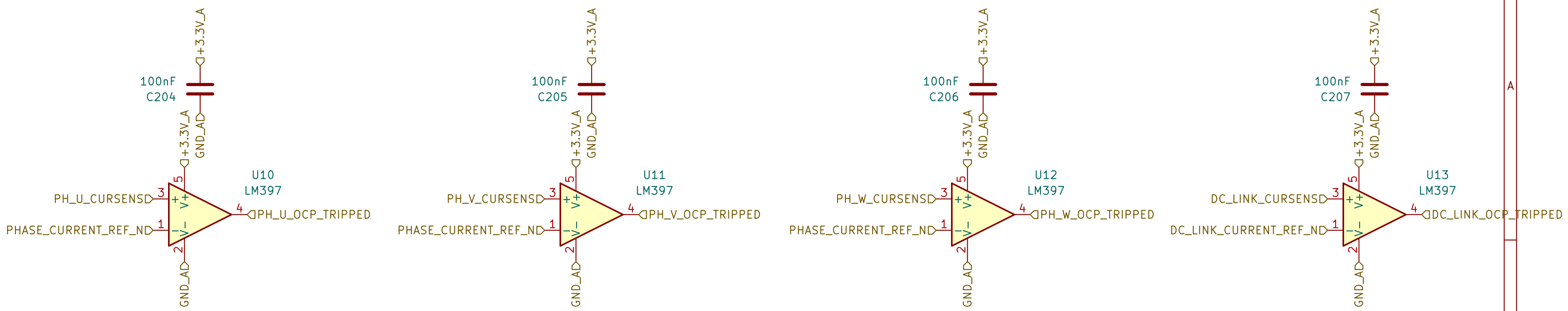
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Control/

File: Control.kicad_sch

***Title: Open Inverter Platform - Control Board***

| | | |
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 8/21 |




Prepared by Thomas Liao

**Corzine Lab, University of California, Santa Cruz**

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

***Title: Open Inverter Platform – Control Board***

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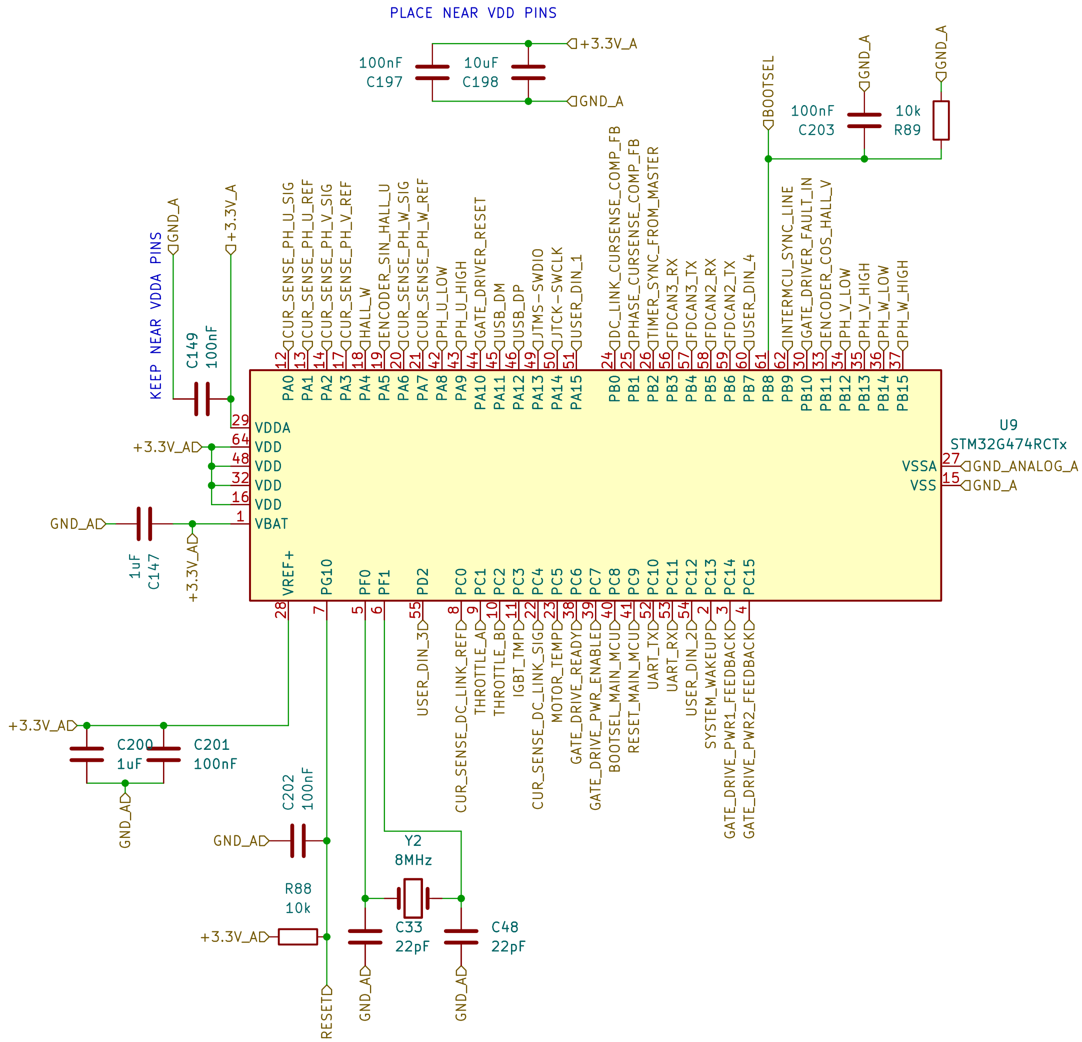
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 10/21 |




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Control/Safety Coprocessor/
File: SafetyCoprocessor.kicad_sch

***Title: Open Inverter Platform — Control Board***

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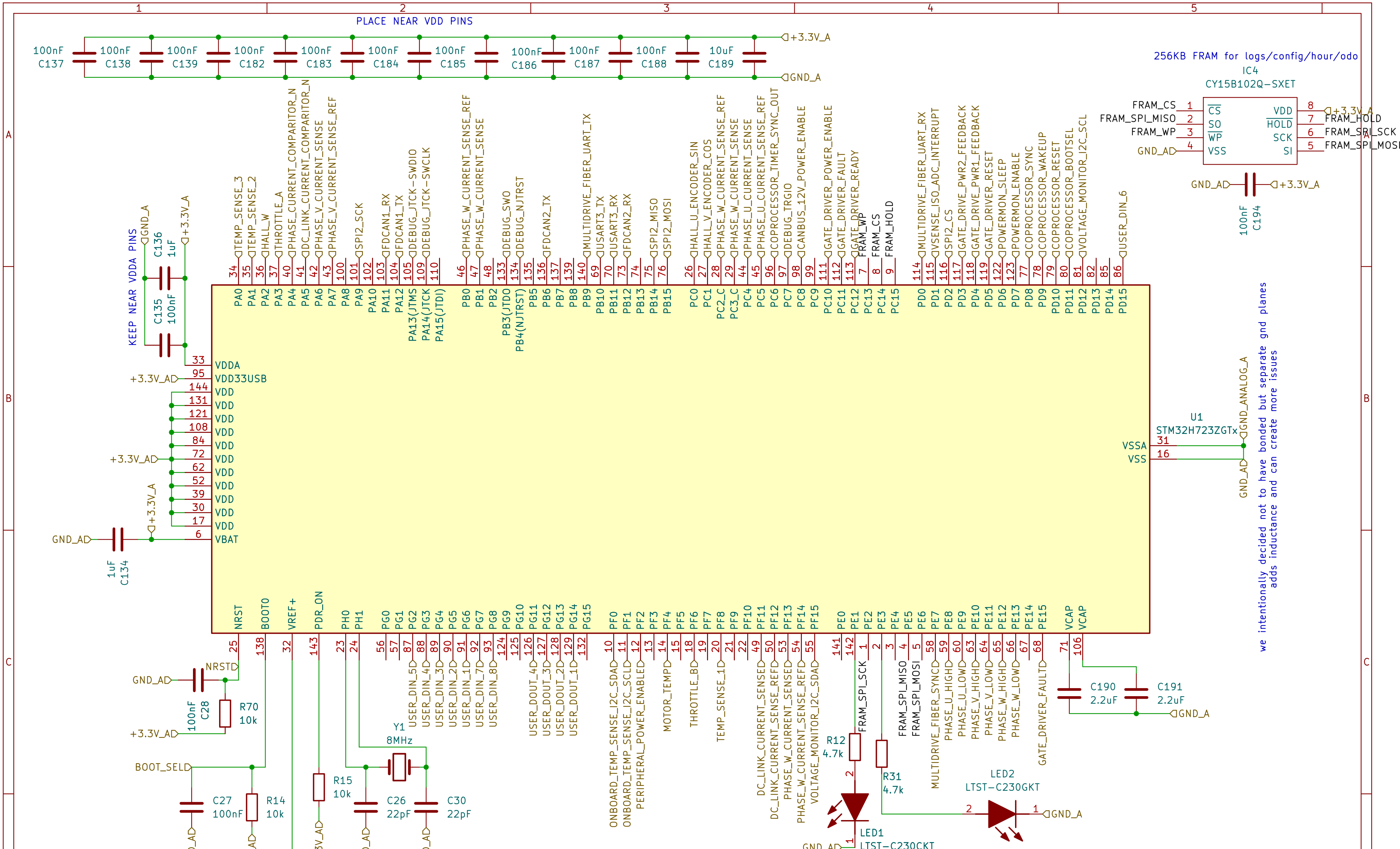
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 16/21 |




Prepared by Thomas Liao

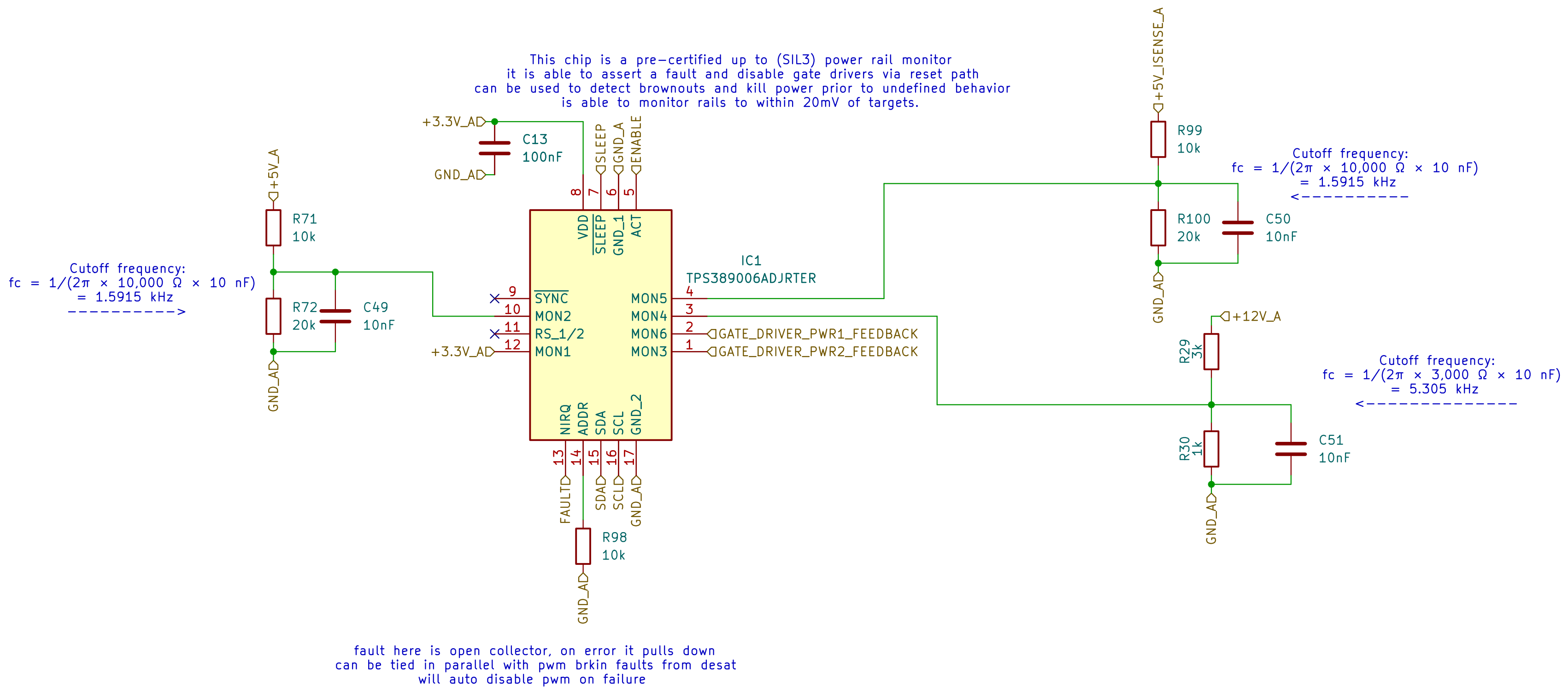
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Control/Main Processor/

File: MainMCU.kicad_sch

***Title: Open Inverter Platform – Control Board***

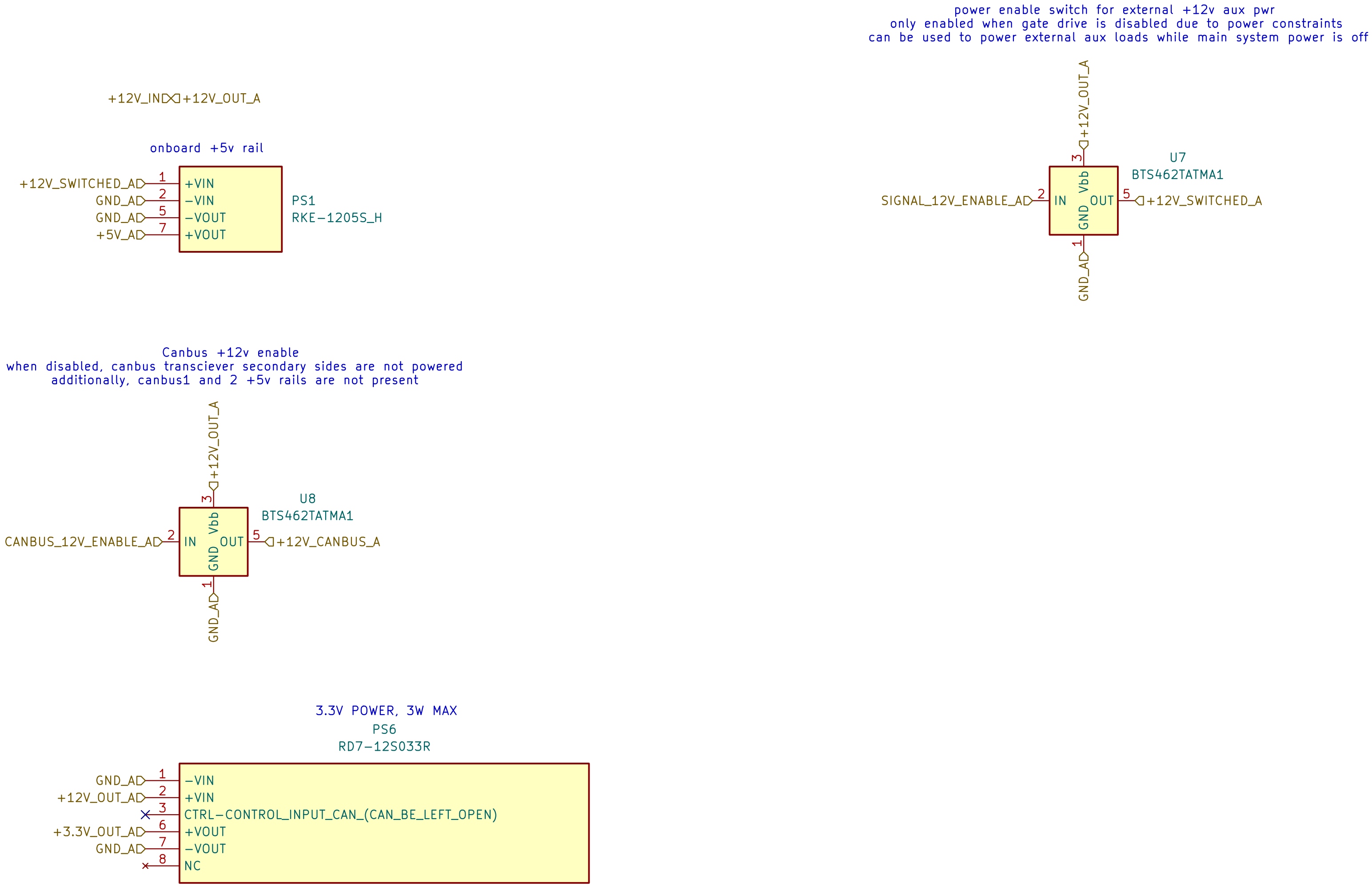
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 19/21 |




Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch

**Title:**

Size: A4 Date: **Rev:**
KiCad E.D.A. 10.0.3 Id: 20/21




Prepared by Thomas Liao

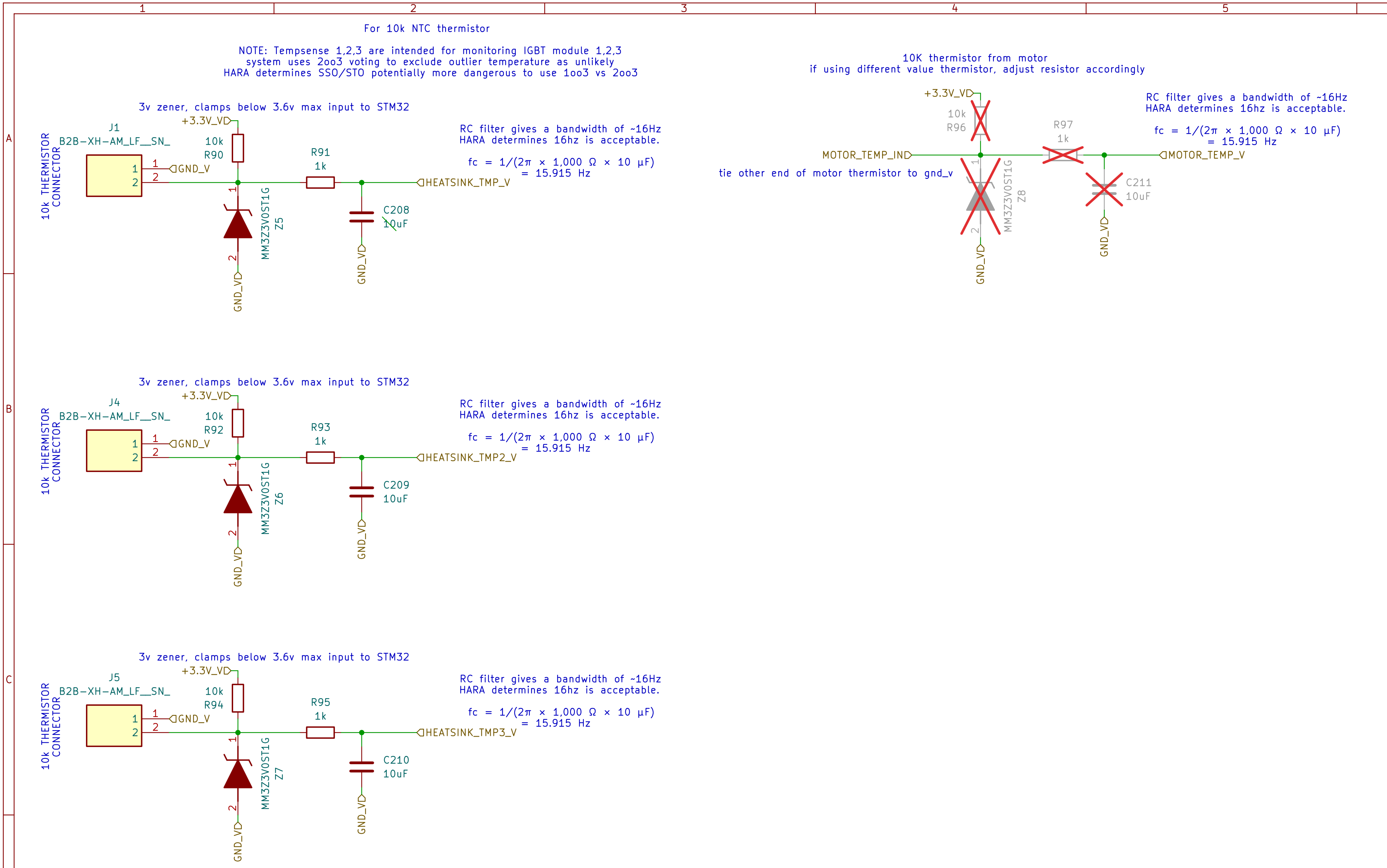
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Power/

File: Power.kicad_sch

***Title: Open Inverter Platform - Control Board***

| | | |
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 9/21 |




Prepared by Thomas Liao

**Corzine Lab, University of California, Santa Cruz**

Sheet: /TemperatureSensors/

File: TemperatureSensors.kicad_sch

***Title: Open Inverter Platform – Control Board***

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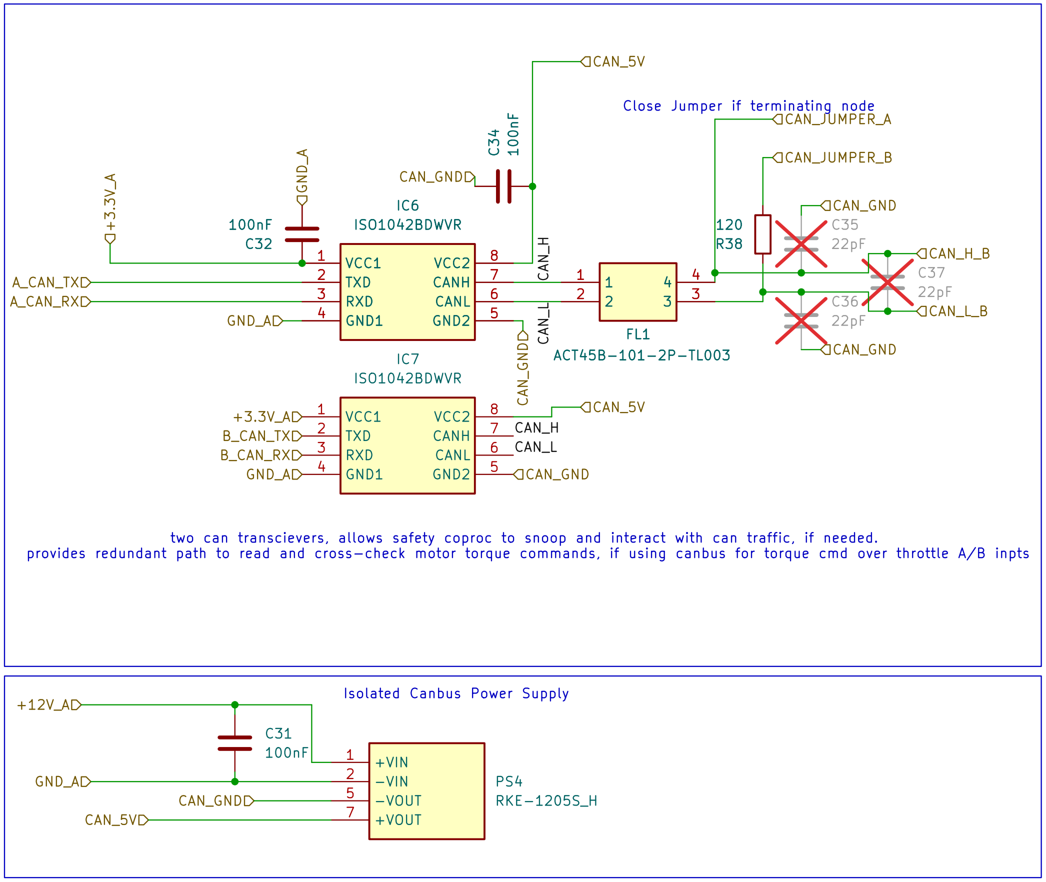
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 9/21 |




Prepared by Thomas Liao

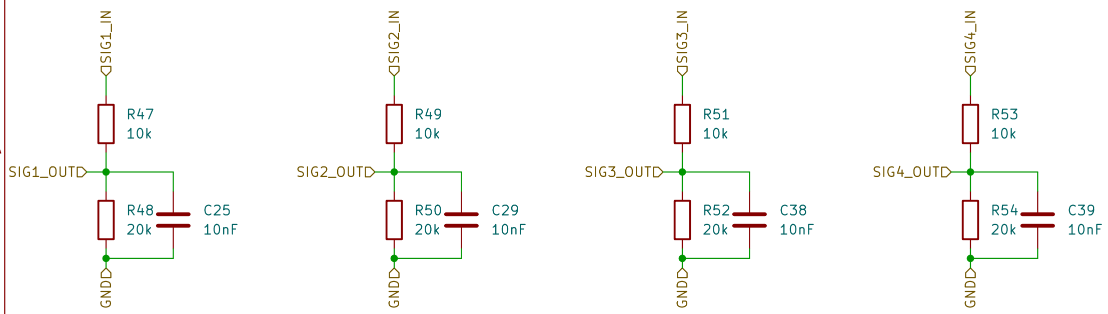
**Corzine Lab, University of California, Santa Cruz**

Sheet: /CAN Transciever/

File: CANTransciever.kicad_sch

***Title: Open Inverter Platform – Control Board***

| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 11/21 |




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

**Title: Open Inverter Platform – Control Board**

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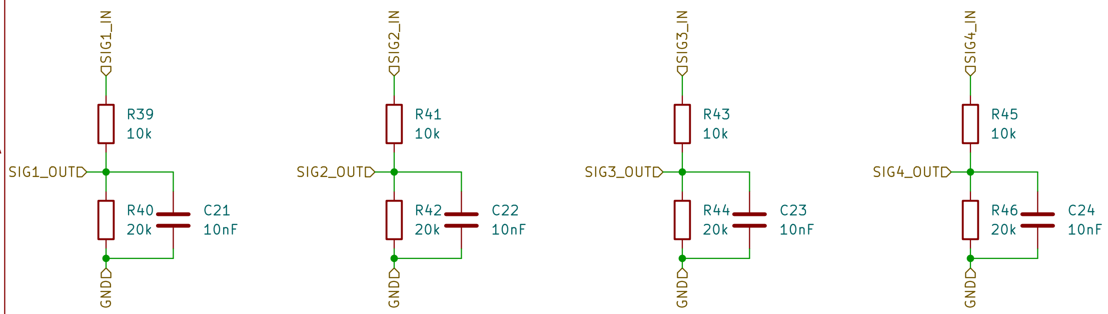
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 12/21 |

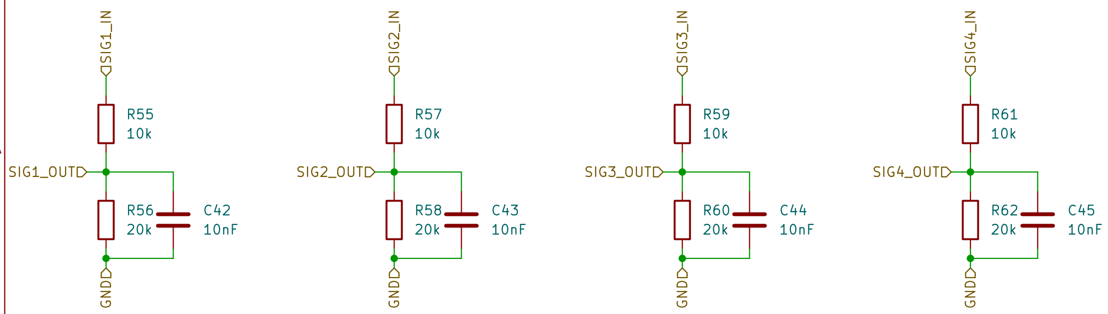



Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch

***Title: Open Inverter Platform – Control Board***

Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 13/21




Prepared by Thomas Liao

**Corzine Lab, University of California, Santa Cruz**

Sheet: /Voltage Divider Network2/

File: VDivNet.kicad_sch

***Title: Open Inverter Platform – Control Board***

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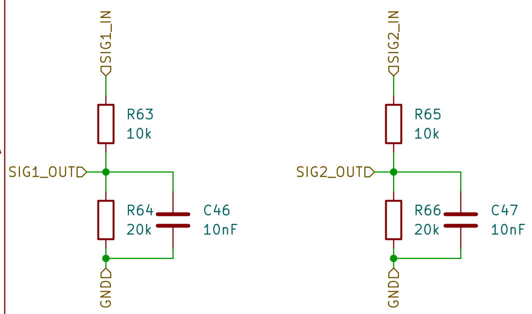
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 14/21 |




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch

***Title: Open Inverter Platform – Control Board***

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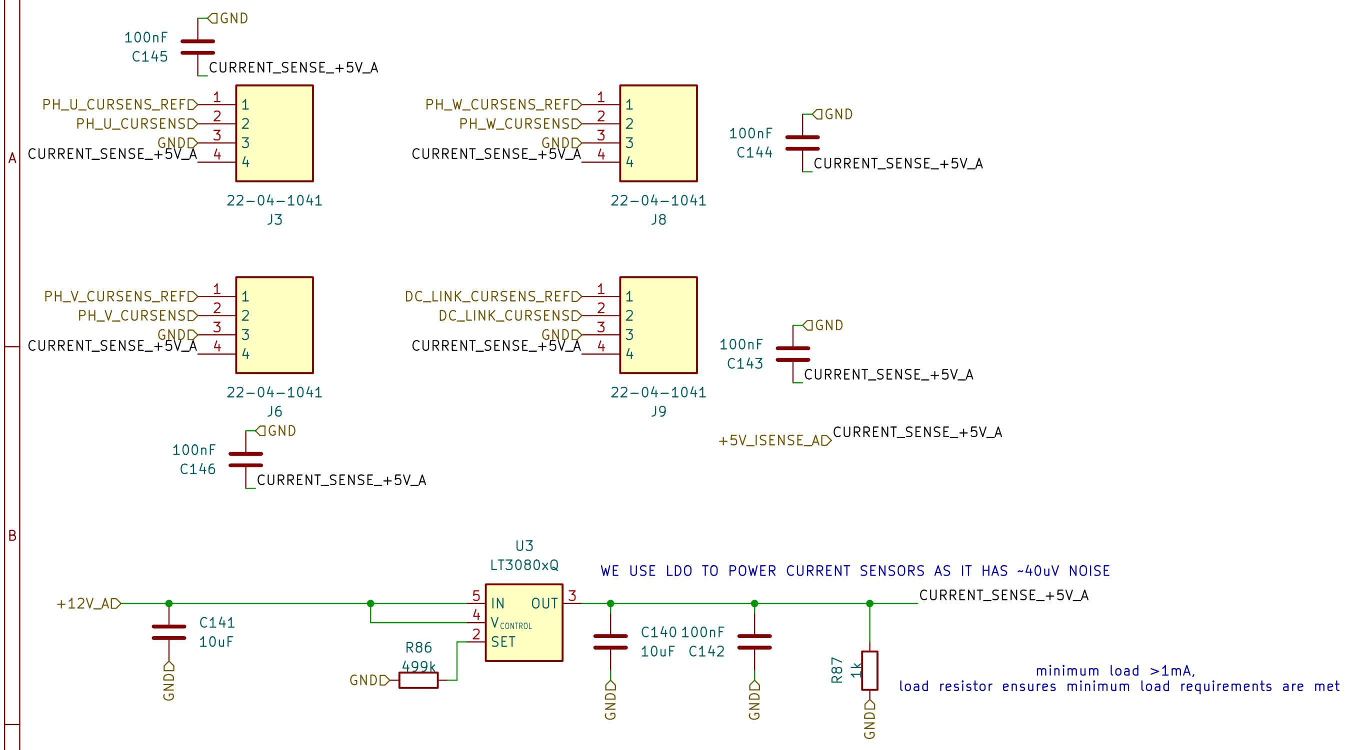
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 15/21 |




Prepared by Thomas Liao
**Corzine Lab, University of California, Santa Cruz**

Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

***Title: Open Inverter Platform – Control Board***

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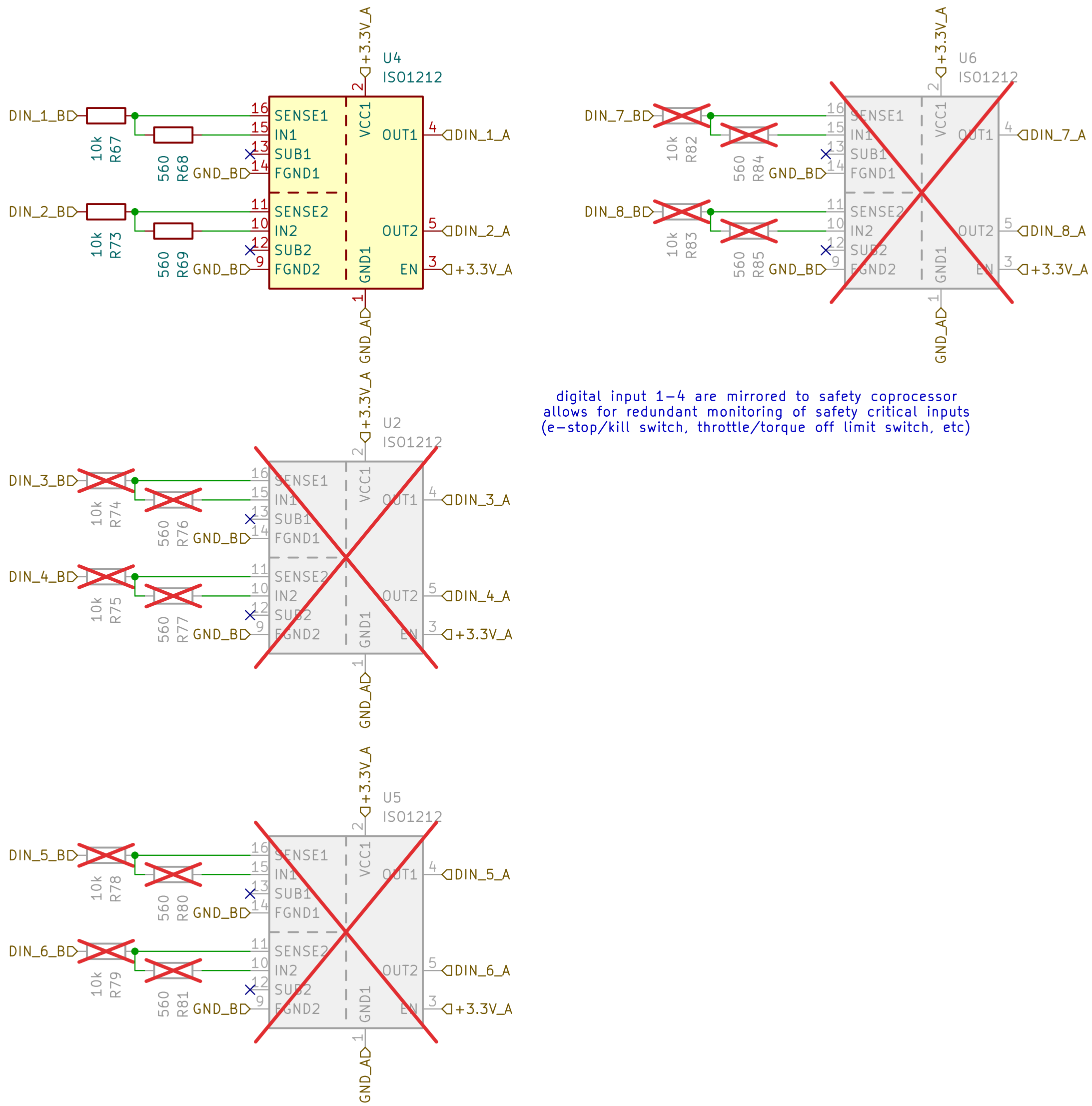
|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 17/21 |




Prepared by Thomas Liao

**Corzine Lab, University of California, Santa Cruz**

Sheet: /Isolated Digital IO/

File: IsolatedDigitalIO.kicad_sch

***Title: Open Inverter Platform – Control Board***

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|---|---|---|
| Size: USLetter | Date: 2026-07-07 | **Rev: G** |
| KiCad E.D.A. 10.0.3 | | Id: 18/21 |